Direct Capital Ratios in State Innovation Funding: Evidence from Pennsylvania and Implications for AI-Era Policy

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Abstract

State governments allocate substantial funding to support entrepreneurship and innovation, typically channeling these dollars through nonprofit intermediary organizations. This paper introduces the Direct Capital Ratio as a metric for evaluating the efficiency of such intermediary models, defined as the proportion of state funding that reaches founders as direct investment capital. Using publicly available IRS Form 990 filings and Federal Audit Clearinghouse data, I analyze Pennsylvania's Ben Franklin Technology Partners network, finding that the Direct Capital Ratio has declined from approximately 42% in FY2023 to 25% in FY2025, with staff compensation now exceeding founder disbursements by over $3 million annually. Comparative analysis of Ohio's JumpStart network reveals similar patterns, suggesting structural rather than organization-specific dynamics. A recent field experiment by Kim, Kim, and Koning (2026) provides causal evidence that AI-enabled startups grow 1.9x faster while requiring 39.5% less external capital, fundamentally challenging the value proposition of high-overhead intermediary models. I propose a policy framework requiring 75% of state innovation funding to flow directly to founders as capital, with the remaining 25% allocated to lightweight coordination functions. This reallocation would increase the number of startups funded, accelerate job creation, and redeploy intermediary talent into the private sector companies that would now have capital to hire them.

Keywords: Innovation Policy; Entrepreneurship; Artificial Intelligence; Nonprofit Intermediaries; State Economic Development; Direct Capital; Ben Franklin Technology Partners

1. Introduction

State governments across the United States invest hundreds of millions of dollars annually in programs designed to foster entrepreneurship and innovation. The dominant model for distributing these funds involves channeling them through nonprofit intermediary organizations that provide a bundle of services: capital investment, mentorship, accelerator programs, networking events, and technical assistance. The underlying assumption is that founders need more than money; they need support systems that intermediaries are uniquely positioned to provide.

This paper challenges that assumption by introducing a simple metric for evaluating the efficiency of intermediary models: the Direct Capital Ratio, defined as the proportion of state funding received by an intermediary that is subsequently disbursed to founders as direct investment capital. Using publicly available data from IRS Form 990 filings and Federal Audit Clearinghouse records, I calculate this ratio for Pennsylvania's Ben Franklin Technology Partners network, one of the oldest and most prominent state-funded innovation programs in the country.

The findings are striking. In the Pittsburgh region, the Direct Capital Ratio has declined from approximately 42% in FY2023 to 25% in FY2025. Staff compensation at the regional intermediary now exceeds total founder disbursements by more than $3 million annually. Comparative analysis of Ohio's JumpStart network reveals a similar pattern, with a Direct Capital Ratio of approximately 21%, suggesting that these dynamics are structural features of the intermediary model rather than artifacts of any particular organization's management.

These findings gain additional significance in light of recent research on artificial intelligence and firm performance. Kim, Kim, and Koning (2026) report results from a field experiment with 515 high-growth startups, finding that firms receiving information about AI deployment grew revenue 1.9x faster while requiring 39.5% less external capital. This suggests that the discovery and guidance functions traditionally provided by intermediaries may now be deliverable through AI-enabled tools at near-zero marginal cost, fundamentally undermining the value proposition of high-overhead intermediary structures.

I propose a policy framework that inverts the current ratio: 75% of state innovation funding should flow directly to founders as capital, with the remaining 25% allocated to lightweight coordination functions. This reallocation would not eliminate intermediary jobs but rather redeploy that talent into the private sector companies that would now have capital to hire. The same state appropriation would fund more companies, accelerate job creation, and generate greater tax revenue returns.

2. Background: The Intermediary Model in State Innovation Policy

The use of nonprofit intermediaries to distribute state innovation funding emerged in the 1980s as policymakers sought to replicate the venture capital ecosystem that had developed in Silicon Valley and Route 128. Pennsylvania's Ben Franklin Technology Partnership, established in 1983, became a model for this approach, creating four regional nonprofit centers tasked with supporting technology-based economic development across the state.

The intermediary model rests on several assumptions. First, that early-stage founders lack access to capital and cannot efficiently connect with investors on their own. Second, that founders benefit from bundled services including mentorship, networking, and technical assistance that pure capital providers do not offer. Third, that nonprofit organizations, governed by mission rather than profit motive, are better positioned than for-profit entities to serve the public interest in economic development.

These assumptions were plausible in an era when information about best practices in entrepreneurship was scarce, investor networks were geographically concentrated, and the transaction costs of connecting founders with resources were high. The question this paper raises is whether these assumptions remain valid in an era of ubiquitous information, AI-enabled discovery tools, and dramatically reduced transaction costs.

The Ben Franklin Technology Partners network currently comprises four regional centers: Ben Franklin Technology Partners of Southeastern Pennsylvania, Ben Franklin Technology Partners of Northeastern Pennsylvania, Ben Franklin Technology Partners of Central and Northern Pennsylvania, and Innovation Works (serving southwestern Pennsylvania). Each center receives appropriations from the Pennsylvania Department of Community and Economic Development (DCED) and supplements this funding with federal grants, private contributions, and investment returns.

3. Data and Methods

3.1 Data Sources

This analysis relies on three primary data sources, all publicly available. First, IRS Form 990 filings provide detailed information on nonprofit revenues, expenses, compensation, and program activities. These filings are accessible through ProPublica's Nonprofit Explorer and the IRS Tax Exempt Organization Search. Second, Federal Audit Clearinghouse records provide audited consolidated financial statements for organizations receiving federal funding, including detailed breakdowns of funding sources and expenditure categories. Third, annual reports published by the intermediary organizations themselves provide supplementary context, though I rely primarily on audited financial data rather than self-reported metrics.

3.2 Defining the Direct Capital Ratio

The Direct Capital Ratio is calculated as:

Direct Capital Ratio = (Dollars Disbursed Directly to Founders as Capital) / (Total State Innovation Funding Received)

The numerator includes equity investments, convertible notes, and direct grants to founders or their companies. It excludes expenditures on staff salaries, program operations, facilities, events, marketing, and other overhead. The denominator includes state appropriations designated for innovation support, excluding federal passthrough funds with different intended purposes (such as Economic Development Administration grants designated for infrastructure rather than direct founder support).

This metric is intentionally simple and conservative. It does not attempt to measure the total value created by intermediaries, which might include network effects, knowledge spillovers, or other difficult-to-quantify benefits. Rather, it measures the efficiency of capital transmission: for every dollar the state intends to support innovation, how much actually reaches the founders doing the innovating?

3.3 Funding Sources

Understanding where innovation funding originates strengthens the accountability case for measuring how it is spent. In both Pennsylvania and Ohio, state innovation funding derives from dedicated revenue mechanisms with specific public policy justifications, not general fund appropriations that might be dismissed as routine government inefficiency.

In Pennsylvania, the Innovate in PA program was established by Act 52 of 2013, authorizing the Department of Community and Economic Development to sell up to $100 million in deferred insurance premium tax credits to insurance companies doing business in the Commonwealth. In July 2015, DCED auctioned these credits to 29 qualified insurance companies at 87 cents on the dollar, generating approximately $87 million in proceeds. These funds flow to the four regional Ben Franklin Technology Partners centers, the Life Sciences Greenhouses, and the Ben Franklin Technology Development Authority's Venture Investment Program. The mechanism is elegant: insurance companies receive credits against their Pennsylvania insurance premium tax liability, and the state receives upfront capital earmarked for innovation. When 75 cents of every dollar raised through this mechanism goes to intermediary overhead before reaching founders, the insurance companies paying into the system are not receiving the economic development return the program promised.

In Ohio, the Third Frontier program was launched in 2002 as a $1.6 billion initiative funded through a combination of general obligation bonds, state general fund appropriations, and proceeds from the Tobacco Master Settlement Agreement. Ohio voters renewed the program in 2010, authorizing an additional $700 million in bonds. The tobacco settlement connection is significant:

Ohio securitized a portion of its MSA payments through the Buckeye Tobacco Settlement Financing Authority, using the proceeds to fund schools and higher education facilities, which freed general fund capacity for Third Frontier investments. JumpStart, the Cleveland-based intermediary analyzed in this paper, receives grants through the Third Frontier Entrepreneurial Services Provider program, most recently $39 million in 2022. When the Direct Capital Ratio at organizations like JumpStart falls below 25%, Ohio taxpayers servicing those bonds are paying for intermediary overhead rather than founder capital.

The funding source matters because it establishes the public's legitimate interest in how the money is spent. These are not charitable donations subject to donor intent. They are public revenues raised through specific mechanisms for specific purposes. Insurance premium tax credits exist because the state promised economic development. General obligation bonds exist because voters approved them for innovation investment. When the majority of these funds are absorbed by the organizations distributing them rather than reaching the founders they were meant to support, the public has standing to demand structural reform.

4. Findings: Pennsylvania's Ben Franklin Network

4.1 Innovation Works (Southwestern Pennsylvania)

Innovation Works serves the Pittsburgh region and southwestern Pennsylvania. Analysis of Federal Audit Clearinghouse filings reveals a declining Direct Capital Ratio over the past three fiscal years:

Fiscal Year	Direct Capital Ratio	Trend
FY2023	~42%	Baseline
FY2024	~37%	-5 percentage points
FY2025	~25%	-12 percentage points

The most recent fiscal year data reveals that staff compensation at Innovation Works now exceeds total founder disbursements by approximately $3 million. This inversion represents a structural shift: the organization spends more on its own personnel than it provides to the founders it ostensibly exists to serve.

It is important to emphasize that this finding does not imply malfeasance or mismanagement. The individuals working at Innovation Works are, by available evidence, competent professionals operating within the constraints of an organizational model that has evolved over four decades. The issue is structural, not personal: the intermediary model creates overhead that consumes an increasing share of available resources.

5. Comparative Evidence: Ohio's JumpStart

To assess whether the patterns observed in Pennsylvania are specific to that state's implementation or reflect broader dynamics of the intermediary model, I conducted parallel analysis of JumpStart, Ohio's primary nonprofit innovation intermediary serving the Cleveland/Northeast Ohio region.

JumpStart operates a similar model to Innovation Works: it receives state funding, provides capital and services to startups, and employs professional staff to manage programs and investments. Initial analysis of JumpStart's publicly available financial filings suggests a Direct Capital Ratio of approximately 21%, even lower than Innovation Works' current ratio.

However, a fair reading of JumpStart's financials requires acknowledging that approximately $11 million reported as consultant expenses may represent services provided directly to founders, including credits for legal services, accounting, and other professional support. If these expenditures are properly categorized as founder-directed resources rather than organizational overhead, JumpStart's effective Direct Capital Ratio would be meaningfully higher than the initial calculation suggests.

This ambiguity points to a broader methodological challenge: the distinction between overhead and founder-directed spending is not always clear in nonprofit financial reporting. Services bundled with capital investment may genuinely benefit founders even if they do not appear as direct disbursements. Future research should develop more granular frameworks for categorizing intermediary expenditures. Nevertheless, even with generous assumptions about founder-directed consulting, the cross-state comparison suggests that intermediary models consume substantial resources before they reach founders in any form.

6. New Evidence: AI and Firm Performance

The preceding analysis documents inefficiency in the current system. A natural response might be that this inefficiency is the necessary cost of providing valuable services that founders cannot obtain elsewhere. Recent research challenges this assumption directly.

Kim, Kim, and Koning (2026) report results from a preregistered field experiment conducted with 515 high-growth startups in a three-month accelerator program. The treatment condition was remarkably simple: firms received structured information about how other companies had successfully integrated AI into their production processes. The control group received standard accelerator programming without this AI-focused content.

The results were substantial:

1. Treated firms discovered 44% more AI use cases, concentrated in product development and strategy.
2. Treated firms completed 12% more tasks.

3. Treated firms were 18% more likely to acquire paying customers.
4. Treated firms generated 1.9x higher revenue.
5. Despite faster growth, treated firms' demand for external capital fell by 39.5%.

The fifth finding is particularly significant for this analysis. The traditional justification for intermediary overhead is that founders need support services to grow effectively. But if AI-enabled information transfer allows founders to grow nearly twice as fast while needing 40% less capital, then the discovery and guidance functions of intermediaries may no longer justify their cost.

The bottleneck, Kim et al. argue, was never access to programs or mentorship. It was discovery: understanding where and how to deploy AI in production. AI itself solves discovery problems. The technology that intermediaries might help founders adopt is the same technology that makes the intermediary model obsolete.

7. Policy Implications: The 75% Direct Capital Mandate

Based on the evidence presented, I propose a straightforward policy reform: state contracts with innovation intermediaries should require that at least 75% of state funding be disbursed directly to founders as capital, with the remaining 25% allocated to lightweight coordination functions.

7.1 Rationale for the 75% Threshold

The 75% threshold represents an inversion of the current observed ratio, where approximately 75% of funding goes to overhead and 25% to founders. This is not arbitrary but rather reflects the empirical finding that most intermediary functions can now be performed at dramatically lower cost through AI-enabled tools and platforms.

A 75% Direct Capital Ratio would mean that for every $10 million in state innovation funding, $7.5 million would reach founders as investment capital and $2.5 million would cover coordination costs. Under current ratios, only $2.5 million reaches founders while $7.5 million covers overhead. The policy simply redirects existing appropriations to their stated purpose.

7.2 Implementation Mechanism

Implementation would require a single modification to DCED contracts with Ben Franklin Technology Partners regional centers: the addition of a Direct Capital Ratio floor. Intermediaries would retain flexibility in how they structure their operations, but would be required to demonstrate that at least 75% of state funds reached founders as capital.

Compliance would be straightforward to verify using the same audited financial data analyzed in this paper. The Federal Audit Clearinghouse already requires detailed reporting of fund flows. Adding a Direct Capital Ratio requirement would create accountability without imposing new reporting burdens.

7.3 Workforce Implications

A common objection to efficiency reforms is that they eliminate jobs. This objection misunderstands the dynamics at play. The intermediary workforce is not eliminated under this proposal; it is redeployed.

When more capital flows to founders, those founders hire. A program officer currently earning $68,000 at an intermediary could become a Chief of Staff at a Series A startup earning $110,000 plus equity. A communications director writing foundation reports could lead marketing at a venture-backed company with stock options. The nonprofit sector does not pay well; it compensates with mission alignment. But mission alignment does not compound. Equity does.

The same talented individuals currently staffing intermediaries would find employment in the expanded startup ecosystem that additional direct capital would create. They would earn more, build wealth through equity participation, and contribute to companies generating tax revenue rather than consuming grant funding.

8. Conclusion

This paper introduces the Direct Capital Ratio as a metric for evaluating the efficiency of state innovation funding intermediaries. Analysis of Pennsylvania's Ben Franklin Technology Partners network reveals that approximately 25% of state funding currently reaches founders as direct capital, with the remainder consumed by intermediary operations. Comparative analysis of Ohio's JumpStart suggests this pattern is structural rather than organization-specific.

Recent field experiment evidence demonstrates that AI-enabled startups grow nearly twice as fast while requiring 40% less external capital. This finding challenges the fundamental value proposition of high-overhead intermediary models: if the discovery and guidance functions they provide can be delivered through AI at near-zero marginal cost, then the overhead they impose is no longer justified.

I propose a 75% Direct Capital Ratio mandate for state innovation funding contracts. This reform would redirect existing appropriations to their stated purpose, increase the number of startups funded, accelerate job creation, and redeploy intermediary talent into private sector companies with greater compensation and wealth-building potential.

The nonprofit industrial complex that has grown around innovation funding is not corrupt. The individuals working within it are not malicious. The structure is simply obsolete. AI has changed the economics of discovery and guidance. Policy should change accordingly.

The relevant question for policymakers is not whether to protect intermediary jobs. It is whether to continue allocating 75 cents of every innovation dollar to overhead, or to flip that ratio and let

the capital compound in the hands of founders building companies, creating jobs, and generating the tax revenue that funds public services.

The math is clear. The evidence is public. The policy change required is a single line in a contract.

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